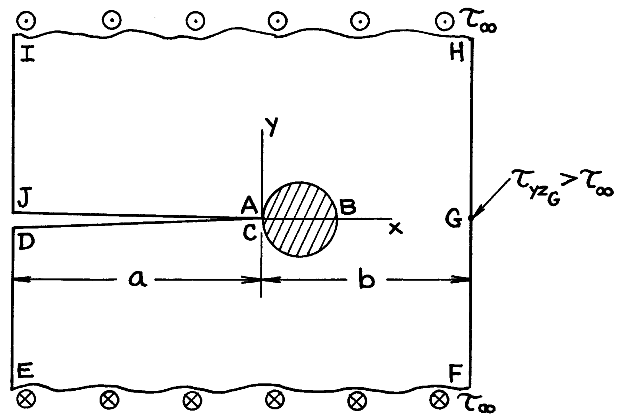
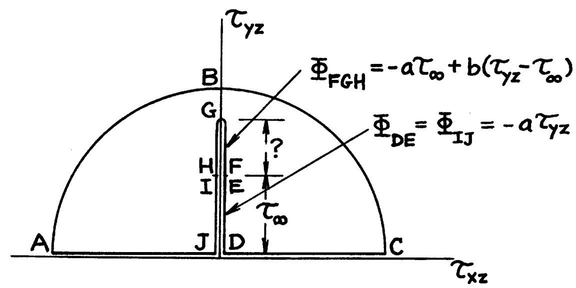
Physical Plane



Stress Plane



Sketch of Stress Function Surface

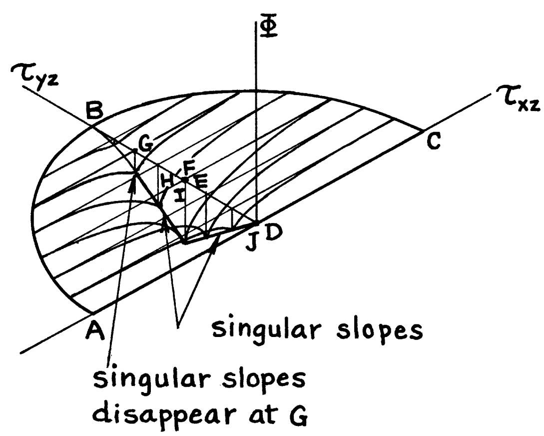
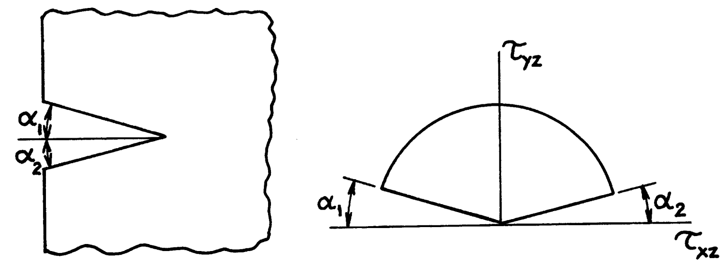
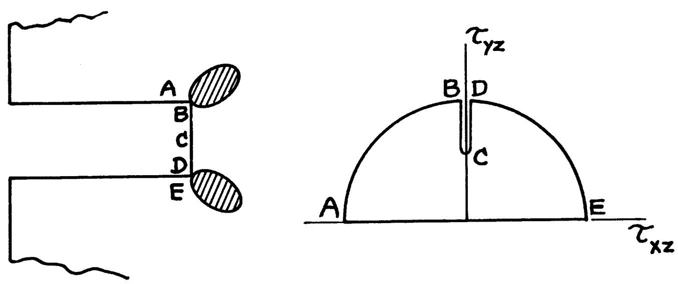


Fig. K.6

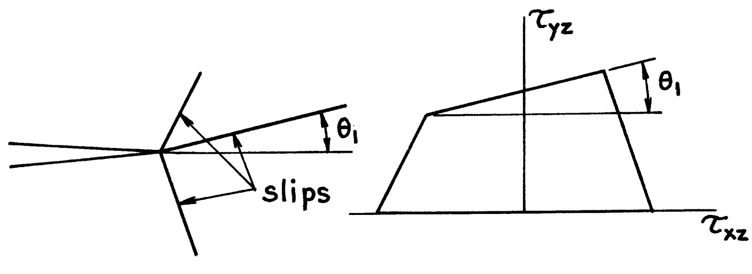
Effect of Angled Sharp Notches:



Effect of Square Ended Notches:



Effect of Discrete Limited Slip Directions:



For Plane Anisotropic Elastic Behavior:

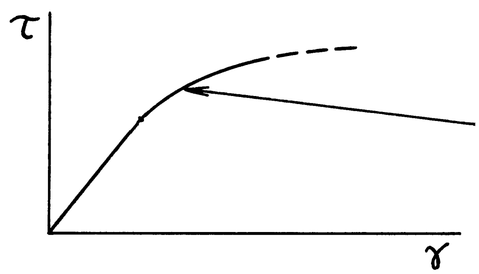
$$\tau_{xz} = A_{xx}\gamma_{xz} + A_{xy}\gamma_{yz}$$

$$\tau_{yz} = A_{yx}\gamma_{xz} + A_{yy}\gamma_{yz} \quad \text{with} \quad \mathrm{A}_{yx} = A_{xy}$$

results in replacing Laplace's Equation for the elastic region by

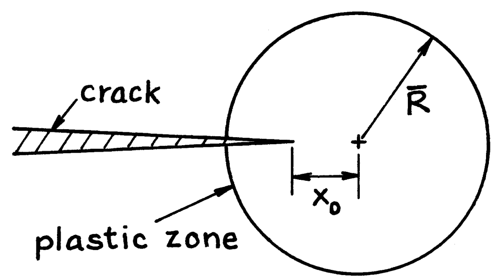
$$A_{xx}\frac{\partial^2 \Phi}{\partial \tau_{xz}^2} + 2A_{xy}\frac{\partial^2 \Phi}{\partial \tau_{xz}\partial \tau_{yz}} + A_{yy}\frac{\partial^2 \Phi}{\partial \tau_{yz}^2} = 0$$

For Power Hardening Plasticity Beyond an Elastic Range:



$$\frac{\tau}{\tau_0} = \left(\frac{\gamma}{\gamma_0}\right)^N$$

$$x_0 = \frac{1-N}{1+N}\overline{R}$$



(See **Rice 1968c** for details and numerical solutions.)

APPENDIX L

TABLE OF COMPLETE ELLIPTIC INTEGRALS, $K(k)$ AND $E(k)$

$k = \sin \alpha$

α°	K	E	α°	K	E	α°	K	E
0	1.5708	1.5708	30	1.6858	1.4675	60	2.1565	1.2111
1	1.5709	1.5707	31	1.6941	1.4608	61	2.1842	1.2015
2	1.5713	1.5703	32	1.7028	1.4539	62	2.2132	1.1920
3	1.5719	1.5697	33	1.7119	1.4469	63	2.2435	1.1826
4	1.5727	1.5689	34	1.7214	1.4397	64	2.2754	1.1732
5	1.5738	1.5678	35	1.7312	1.4323	65	2.3088	1.1638
6	1.5751	1.5665	36	1.7415	1.4248	66	2.3439	1.1545
7	1.5767	1.5649	37	1.7522	1.4171	67	2.3809	1.1453
8	1.5785	1.5632	38	1.7633	1.4092	68	2.4198	1.1362
9	1.5805	1.5611	39	1.7748	1.4013	69	2.4610	1.1272
10	1.5828	1.5589	40	1.7868	1.3931	70	2.5046	1.1184
11	1.5854	1.5564	41	1.7992	1.3849	71	2.5507	1.1096
12	1.5882	1.5537	42	1.8122	1.3765	72	2.5998	1.1011
13	1.5913	1.5507	43	1.8256	1.3680	73	2.6521	1.0927
14	1.5946	1.5476	44	1.8396	1.3594	74	2.7081	1.0844
15	1.5981	1.5442	45	1.8541	1.3506	75	2.7681	1.0764
16	1.6020	1.5405	46	1.8691	1.3418	76	2.8327	1.0686
17	1.6061	1.5367	47	1.8848	1.3329	77	2.9026	1.0611
18	1.6105	1.5326	48	1.9011	1.3238	78	2.9786	1.0538
19	1.6151	1.5283	49	1.9180	1.3147	79	3.0617	1.0468
20	1.6200	1.5238	50	1.9356	1.3055	80	3.1534	1.0401
21	1.6252	1.5191	51	1.9539	1.2963	81	3.2553	1.0338
22	1.6307	1.5141	52	1.9729	1.2870	82	3.3699	1.0278
23	1.6365	1.5090	53	1.9927	1.2776	83	3.5004	1.0223
24	1.6426	1.5037	54	2.0133	1.2681	84	3.6519	1.0172
25	1.6490	1.4981	55	2.0347	1.2587	85	3.8317	1.0127
26	1.6557	1.4924	56	2.0571	1.2492	86	4.0528	1.0086
27	1.6627	1.4864	57	2.0804	1.2397	87	4.3387	1.0053
28	1.6701	1.4803	58	2.1047	1.2301	88	4.7427	1.0026
29	1.6777	1.4740	59	2.1300	1.2206	89	5.4349	1.0008
30	1.6858	1.4675	60	2.1565	1.2111	90	∞	1.0000

The definitions of the complete elliptic integrals of the first kind, $K(k)$, and the second kind, $E(k)$, are as follows:

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

and

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \quad d\varphi$$

Approximate Formulas for $E(k); k = \sin\alpha$

$$E(k) = \left[1 + 1.464\,(\cos\alpha)^{1.65}\right]^{1/2}$$

Accuracy: 0.1%
References: **Rawe 1970**; see also **Merkle 1973, Raju 1979**

$$E(k) = \frac{\pi}{2} \cdot \frac{1}{1+\lambda} \cdot \frac{4 - 0.18\lambda^4}{4 - \lambda^2}$$

where $\lambda = \tan^2 \frac{\alpha}{2}$

Accuracy: 0.05%
Reference: **Tada 2000**

APPENDIX M

A TABLE OF GAMMA FUNCTION $\Gamma(x)$

x	$\Gamma(x)$	x	$\Gamma(x)$
1.000	1.00000	1.500	.88623
1.025	.98617	1.525	.88729
1.050	.97350	1.550	.88887
1.075	.96191	1.575	.89094
1.100	.95135	1.600	.89351
1.125	.94174	1.625	.89657
1.150	.93304	1.650	.90012
1.175	.92520	1.675	.90414
1.200	.91817	1.700	.90864
1.225	.91192	1.725	.91361
1.250	.90640	1.750	.91906
1.275	.90160	1.775	.92499
1.300	.89747	1.800	.93138
1.325	.89400	1.825	.93825
1.350	.89115	1.850	.94561
1.375	.88891	1.875	.95345
1.400	.88726	1.900	.96177
1.425	.88618	1.925	.97058
1.450	.88566	1.950	.97988
1.475	.88567	1.975	.98969
1.500	.88623	2.000	1.00000

Values of $\Gamma(x)$ for $x<1$ and $x>2$ are calculated by the following formula:

$$\Gamma(x) = \frac{\Gamma(x+1)}{x} \text{ or } \Gamma(x) = (x-1)\Gamma(x-1)$$

Examples:

1) $$\Gamma(0.65) = \frac{\Gamma(1.65)}{0.65} = \frac{0.90012}{0.65} = 1.38480$$

2) $$\begin{aligned}\Gamma(3.65) &= 2.65 \text{ x } \Gamma\ (2.65) \\ &= 2.65 \text{ x } 1.65 \text{ x } \Gamma\ (1.65) \\ &= 2.65 \text{ x } 1.65 \text{ x } 0.90012 = 3.93577\end{aligned}$$

PROPERTIES OF GAMMA FUNCTION $\Gamma(x)$

Definition:

$$\Gamma(x) = \begin{cases} \displaystyle\int_0^\infty e^{-t} t^{x-1} dt & \text{for } x \geq 0^{*} \\ \displaystyle\lim_{n\to\infty} \frac{n! n^{x-1}}{x(x+1)(x+2)\cdots\cdots(x+n-1)} & \text{for any } x^{**} \end{cases}$$

* when x is complex; $Re(x)>0$
** real or complex

Basic Formulas:

$$\Gamma(x+1) = x\Gamma(x)$$

$$\Gamma(n) = (n-1)! \quad (\text{ n = positive integer})$$

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

$$\Gamma(x)\Gamma\left(x + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2x-1}}\Gamma(2x)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \ \Gamma(1) = 1, \ \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

For crack problems, the following formulas are useful.

(1)

$$\int_0^1 x^{2\alpha+1}\left(1-x^2\right)^\beta dx = \left(\frac{1}{2}\int_0^1 x^\alpha (1-x)^\beta dx\right)$$

$$= \int_0^{\pi/2} (\sin\theta)^{2\alpha+1} (\cos\theta)^{2\beta+1} d\theta$$

$$= \frac{\Gamma(\alpha+1)\Gamma(\beta+1)}{2\Gamma(\alpha+\beta+2)} \left(= \frac{1}{2} B(\alpha+1, \ \beta+1)\right)$$

$$\left(= \frac{\alpha!\beta!}{2(\alpha+\beta+1)!} \ \text{when } \alpha, \beta = \text{ positive integers}\right)$$

(2) Special cases: $2\alpha + 1 = \gamma,\ \beta = -\frac{1}{2}$

or $\alpha = -\frac{1}{2},\ 2\beta + 1 = \gamma$

$$\int_0^{\pi/2} (\sin\theta)^{\gamma} d\theta = \int_0^{\pi/2} (\cos\theta)^{\gamma} d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+1\right)}$$

when $\gamma = n$ (n = positive integer)

$$\begin{cases} = \dfrac{(n-1)(n-3)(n-5)\cdots\cdots 3.1}{n\,(n-2)(n-4)\cdots\cdots 4.2} \cdot \dfrac{\pi}{2} & (n = \text{ even}) \\[2ex] = \dfrac{(n-1)(n-3)(n-5)\cdots\cdots 4.2}{n\,(n-2)(n-4)\cdots\cdots 3.1} & (n = \text{ odd}) \end{cases}$$

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Reference Index

U

V

W

Y

Z

SUBJECT INDEX

T

U

FREE SOFTWARE (SMARTCRACK-LITE)

Free software (SmartCrack-Lite) by Engineering Mechanics Technology, Inc. is available to purchasers of this book as a supplement to this Handbook upon request. A description of this software can be found on the following page.

This software is not an ASME product, but by special arrangement with the developers, purchasers of this book can request free copies by contacting:

Engineering Mechanics Technology, Inc.
4340 Stevens Creek Blvd., Suite 166
San Jose, CA 95129
(408) 247-9274 (PHONE)
(408) 247-9272 (FAX)
dharris@emtinc.com (E-MAIL)
www.emtinc.com (WEB ADDRESS)

Proof of purchase, a copy of the invoice or packing slip that comes with the book, will be required to obtain the software. This software is owned by Engineering Mechanics Technology, Inc.

(This is not an endorsement by ASME, implied, deliberate or otherwise, of the software (SmartCrack-Lite) or of Engineering Mechanics Technology, Inc. and should not be construed as such.)

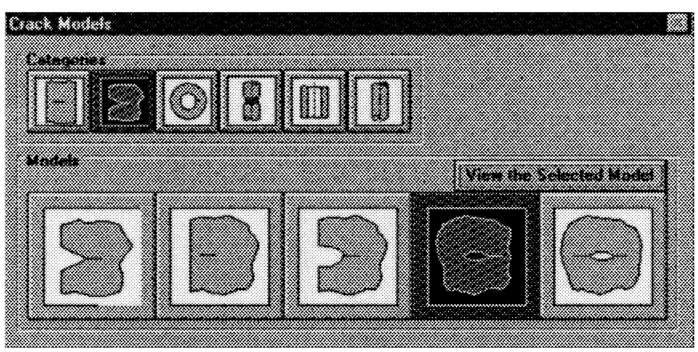
SOFTWARE FOR EVALUATION OF STRESS INTENSITY FACTORS BY USE OF FORMULAS IN THE STRESS ANALYSIS OF CRACKS HANDBOOK

The third edition of *The Stress Analysis of Cracks Handbook* contains many formulas and plots for evaluation of stress intensity factors. The use of such results is made more convenient by the software that is included in the accompanying computer diskette. The software is called SmartCrack-Lite, is in Windows, and contains the following 21 K-solutions included in the Handbook.

Standard Specimens	Infinite & Semi-Infinite	Round Bars
center cracked strip	crack from V-notch	exterior circum. crack
double edge crack	crack near edge	interior crack
single edge crack	crack from elliptical notch	**Hollow Round Bars**
3-point bend	crack from elliptical hole	interior crack
compact tension	2 cracks from ell. hole	exterior crack
arc shaped	**Disks**	**Shells**
round compact	edge loaded center crack	axial crack cylinder
	edge crack	circum. crack cylinder
		sphere

Many of the K-solutions include the influence function, which normally requires numerical integration for their use. The numerical integration procedure is included in the software and is transparent to the user. This makes application of the influence functions much easier. Further description of the software is included on the accompanying diskette, which also describes the installation procedure for SmartCrack-Lite. The following are the minimum system requirements to run SmartCrack-Lite: any IBM-compatible running Windows 3.1, 95, 98 or NT 4.0, a hard disk, and a 3 1/2" floppy disk.

The dialog box for the selection of cracks in the "infinite and semi-infinite planes" category is shown below as an example.



The K-solutions available in the software are a subset of the solutions included in a more comprehensive software package called SmartCrack, hence the name SmartCrack-Lite. The full SmartCrack software package, which is available from Engineering Mechanics Technology, Inc., in San Jose, California, contains many additional K solutions, and is able to perform fatigue and stress corrosion crack growth calculations, evaluate critical crack lengths, and evaluate crack opening areas. An extensive compilation of material fatigue crack growth characteristics is included.